MARMARA UNIVERSITY - Faculty of Engineering

Computer Science Engineering

SYLLABUS

2020-2021 Spring Semester

Course Code	Course Name	Course Type	Weekly Course Hours T	A	L	Credits	ECTS	Campus / Weekly Time & Classroom Schedule
IE3035	Operations Research	Compulsory	3	0	0	3	5	Monday 9:00-10:50 Tuesday 10:00-10:50 Meeting ID: 918 6185 3076 Passcode: 001122
Prerequisite	MATH2056	Prerequisite to						

Course Lecturer	Assoc. Prof. Ali Fuat ALKAYA	Office Hours Schedule	Monday 13:00-14:00
E-mail	falkaya@marmara.edu.tr		
Phone	(0216) 777 35 32	Office / Room No	Zoom Meeting ID: 754 2298 5847, Passcode: 1357
Teaching Assistant(s)	TBA	Phone	
E-mail		Office / Room No	

Course Description

This is an introductory course in the methods, principles and fundamentals of operations research. Operations Research studies analysis and planning of complex systems. This course will focus on developing mathematical modelling of computer science related problems and application of linear programming techniques. The course includes discussions of the simplex algorithm and other methods to derive solutions for the derived models. The Excel Solver software is also used in the course to solve linear programming problems. Discussions (Sensitivity Analysis) are included as to how changes or variations in a linear programming's parameters affect the optimal solution.

Textbooks and/or References

1. Operations Research, Hamdy A. Taha, Pearson, Any Edition
2. Wayne L. Winston, Operations Research, Cengage Learning US, Any Edition
3. www.canvas.net (for Announcements and file sharing)
4. ues.marmara.edu.tr (for uploading quizzes and exams)

WEEK	Date	TOPICS	Reference No - Section
Week 1	1.03.2021	Introduction to Operations Research	Chapter 1
Week 2	8.03.2021	Linear Programming – Graphical Solution	Chapter 2
Week 3	15.03.2021	Linear Programming – Developing Models	Chapter 2
Week 4	22.03.2021	Linear Programming – Simplex Method	Chapter 3
Week 5	29.03.2021	M Method	Chapter 3
Week 6	5.04.2021	Two Phase Method	Chapter 3
Week 7	12.04.2021	Sensitivity Analysis	Chapter 3
Week 8	26.04.2021	Duality	Chapter 4
Week 9	3.05.2021	Transportation Problem	Chapter 5
Week 10	10.05.2021	Assignment Problem	Chapter 5
Week 11	17.05.2021	Network Model	Chapter 6
Week 12	24.05.2021	Integer Linear Programming	Chapter 9
Week 13	31.05.2021	Travelling Salesman Problem	Chapter 11
Week 14	7.06.2021	Heuristic Programming	Chapter 10

Evaluation Tools

Evaluation Tool	Quantity	Date	Weight in Total (%)	Weight in Semester Evaluation (%)
Final Exam	1		40	0
Final Make-up Exam (if exists)	1		40	0
Semester Evaluation			60	100
Midterm(s)	1		30	50,0
Quizzes	4		20	33,3
Project(s)				
Homeworks (will not be evaluated)	4		0	0,0
Laboratory				
Attendance (through polls)	10		10	16,7

*** *Lifelong Learning Programme (LLP)* ***

Language of Instruction: English

Evaluation Tool	Quantity	Student Workload Hours	Evaluation Tool	Quantity	Student Workload Hours
Theoretical Hours	--	42,0	Applied Hours	--	0,0
Midterm		15,0	Final	1	20,0
Quiz			Project		
Laboratory			Homework	4	40,0
Atelier			Seminar		
Field Study			Presentation		
Other			Self Study		
			TOTAL :	5	117,0

Recommended ECTS Credit (Total Hours / 25) : 5